Fiscal Year:	FY 2005	**Task Last Updated:**	FY 10/27/2005
PI Name:	Bloomfield, Susan A. Ph.D.		
Project Title:	Increasing the Efficiency of Exercise Countermeasures for Bone Loss		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Bone Loss Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sbloom@tamu.edu	**Fax:**	FY 979-862-1692
PI Organization Type:	UNIVERSITY	**Phone:**	979-845-2871
Organization Name:	Texas A&M University		
PI Address 1:	Department of Health & Kinesiology		
PI Address 2:	400 Harvey Mitchell Pkwy, Suite 300		
PI Web Page:			
City:	College Station	**State:**	TX
Zip Code:	77843-4375	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
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No. of Post Docs:	0	**No. of PhD Degrees:**	1
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No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Baldwin, Kenneth (University of California, Irvine) Hogan, Harry (Texas A&M University)		
Grant/Contract No.:	NCC 9-58-BL00404		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The goal of the proposed research is to test the hypothesis that astronaut exercise time designed to mitigate bone loss can be reduced if combined with the appropriate pharmacological agent. The specific objectives are to define an optimal exercise regimen that will minimize bone loss during prolonged unloading and then to test reduced doses of that exercise regimen in combination with alendronate administration. Importantly, skeletal muscle function, mass, and phenotype will be simultaneously tested to assure that any reductions in exercise time do not sacrifice benefits to muscle. These experiments utilize the well-accepted model of hindlimb unloading (HU) in mature adult rats in order to test these concepts in a far more comprehensive way than is possible with human subjects. Specific aim #1 will test two different modes of loading (electrically stimulated muscle contractions and voluntary resistance exercise) for their effects on mitigating bone and muscle decrements during 28 days of HU. Outcome measures include in vivo tests of bone density/geometry and muscle function, and detailed ex vivo studies of bone mechanical properties, histomorphometric variables, and skeletal muscle anabolic/ catabolic state. The regimen judged most successful in mitigating bone loss will be further refined in specific aim #2, by testing varying frequency or session durations during HU. In addition, a dose-response study using alendronate during HU will be performed. That volume of training producing the maximal benefit to bone and acceptable benefit to muscle will then be tested in combination with the minimal dose of alendronate producing significant attenuation of bone loss. Specific aim #3 will test systematically reduced volumes of the optimized loading regimen in combination with alendronate to determine what reduction in exercise time can be achieved without sacrificing benefits to bone and muscle. Our Year 1 activities have focused largely on methodological improvements and the construction of our own version of the flywheel resistance training device for rodents, in consultation with consultant Dr. Jim Fluckey. Following construction of the flywheel device, considerable effort has been invested in perfecting the operant conditioning of our adult rats required before we begin hindlimb unloading experiments using flywheel training as an intervention. Simultaneously experiments are in progress to define the strain magnitude and strain rates engendered by both stimulated muscle contractions, our other form of “resistance training” in these projects. We will proceed to measuring strain on tibial bone during flywheel resistance training in Year 2. Another major accomplishment of Year 1 was to revise our method to provide chronic, repeated muscle stimulation as a resistance training modality. This involved a visit by our Senior Research Associate to the laboratory of collaborator Ken Baldwin at UC-Irvine to learn how to use fine wire electrodes as a replacement for our tibial nerve cuffs, used in our previous NSBRI project. This results in significant time savings by subtracting out the survival surgery required to implant the nerve cuffs, reduced trauma to our animal research subjects, and, importantly, better science results. Previous experimental results suggested that the implanted nerve cuff caused low-grade inflammation resulting in smaller muscles and reduced muscle torque independent of the main experimental treatments. We are rapidly progressing to the key experiments designed to answer Specific Aim #1: which of our two resistance training paradigms provides the best mitigation of bone loss during simulated microgravity while retaining some benefit to lower limb muscle as well. These experiments can be completed in the first 7 months of this funding year, allowing us to proceed to the more detailed dose-response experiments testing various frequencies (training days/week) and durations (e.g. number of contractions) of the “best” regimen selected during Specific Aim #1 experiments. We can also simultaneously do our dose-response experiments with alendronate treatment during hindlimb unloading in order to define the minimum effective dose. These studies are essential, as such dose-response studies have not been performed with this tail suspension model and we cannot presume that doses effective with estrogen deficiency-induced bone loss will also be effective in the context of the more potent catabolic effects of disuse. Once the most effective exercise and alendronate doses have been defined, we can then progress to experiments defining the interactive and possible synergistic effects of the two interventions on bone and muscle. These integrated experiments address five critical questions on the current Critical Path Roadmap relevant to bone health and skeletal muscle function. They will also provide fundamental and unique information about the biological interaction of these two tightly integrated tissues in response to the anabolic stimulus of exercise and the anti-resorptive effects of alendronate. The outcomes of these animal protocols can be easily translated to human studies using appropriate ground- and flight-based test beds.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our detailed dose-response studies of the bone response to resistance training applied as a countermeasure during a period of simulated microgravity will help define the training stimulus required for maintaining bone health during prolonged periods of bed rest in clinical patients or others restricted to bed rest. Subsequent trials will test the efficacy of alendronate, an FDA-approved anti-resorptive agent commmonly prescribed for osteoporotic patients, versus resistance training versus combined alendronate and exercise in maintaining bone health during a period of unloading. These studies will answer whether there is an additive or synergistic effect of these pharmacological and physiological approaches. Given the numbers of Americans currently taking alendronate and other closely related bisphosphonate agents who also exercise, this could prove to be very useful data that could be used effectively to predict the human response to the combined effects of exercise and bisphosphonate treatment. Finally, our experiments testing various durations of exercise in combination with alendronate will quantify the minimum volume of exercise required to achieve the desired outcome of maintaining bone mass during prolonged periods of non-weightbearing when combined with this anti-resorptive treatment. This has important implications for rehabilitation specialists, physical therapists, and any clinicians treating bedridden patients.
Task Progress:	Several critical tasks were accomplished in this first year: A. Several phone conferences with consultant Jeffrey Jones, M.D. (JSC flight surgeon) and all investigators (Bloomfield, Hogan and Ken Baldwin [at UC-Irvine]) to troubleshoot our research design before embarking on full project. B. Building our version of the Arkansas flywheel resistance trainer device: In consultation with our consultant Dr. Jim Fluckey, we “reverse engineered” a version of this device which provides for resistance training (specifically, leg squats) in rats while maintained in the tail suspension posture. This is an important aspect of the model, so that no weightbearing on the hindlimb bone is incurred. Multiple modifications on the original Arkansas design have been made by Hogan graduate students Brent Vyvial (supported on this project) and Justin Alcorn (supported by other funds): • Addition of a brake on the flywheel to assure smooth reversal (essential to assuring rat gets eccentric as well as concentric muscle contractions) • Generation of program language to operate the brake automatically • Addition of an optical sensor • Replace circular pulley on cable between rat and flywheel with cam-shaped pulley • Addition of new load cell in line with cable to ensure more accurate data on forces generated during leg squats • Replacement of ball bearings with needle bearings • Reduction in size of flywheel shaft by half, which allows more rotations of flywheel for given force applied C. Learning how to operantly condition our rats to respond to a light stimulus to perform flywheel resistance training, with consultant Jim Fluckey. D. Revision of methods to provide chronic, repeated muscle stimulation as alternative resistance training mode: Previously we had used an implanted nerve cuff, which required a survival surgery. However, preliminary data indicated possible chronic inflammation producing

	smaller muscles in the nerve cuff-implanted leg. Bloomfield Sr. Research Associate Jan Stallone traveled in February 2005 to co-investigator Ken Baldwin's lab (Irvine, CA) to learn a method of percutaneously placing fine wire electrodes for repetitive protocols, which we have successfully transferred to our full-size (500-600 g) adult rats. This reduces trauma and anesthesia time to the animals and should produce better scientific results. E. Quantification of bone strain with resistance training protocols: In order to achieve this goal we have been developing methods for implanting uniaxial strain gages on the tibial surface during muscle stimulation protocols, first for non-survival protocols. Data is reported in this annual report for some of these preliminary trials. We are developing a survival surgery strain gage implantation for multiple day measurements, with the goal of acquiring strain data from the same ra
Bibliography Type:	Description: (Last Updated: 01/20/2026)
Articles in Peer-reviewed Journals	Narayanan R, Allen MR, Gaddy D, Bloomfield SA, Smith CL, Weigel NL. "Differential skeletal responses of hindlimb unloaded rats on a vitamin D-deficient diet to 1,25-dihydroxyvitamin D3 and its analog, seocalcitol (EB1089)." Bone. 2004 Jul;35(1):134-43. PMID: 15207749 , Jul-2004
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Presentation	Baek, K., A. Currado, M. Allen, S. Bloomfield. "Effect of food restriction and hindlimb unloading on serum leptin and histomorphometric measures of cancellous bone." N/A Jan-2004
Presentation	Bloomfield, S. A., M. R. Allen, K. Baek, J. Swift, and H. A. Hogan "Pharmacological and mechanical interventions to accelerate bone recovery after hindlimb unloading to better match rate of muscle function recovery" N/A Jan-2005
Presentation	Hogan, H. A., W. E. Rogers, MR Allen, SA Bloomfield and CL Smith "Mechanical properties of bone in hindlimb unloaded adult male rats treated with raloxifene and estradiol." N/A Jan-2005
Presentation	Narayanan, R., S. Karmakar, C. Seng, Q. He, L. J. Suva, N. L. Weigel, S. A. Bloomfield and C. L. Smith "Evaluation of Estrogen and Vitamin D Receptor Ligands as Countermeasures for Disuse-Induced Bone Loss in Unloaded and Ambulatory Rats: Are responses to Vitamin D Receptor Ligands Gender-Specific?" N/A Jan-2005
Presentation	Swift, J. M., T. F. Kennedy, S. A. Bloomfield "Intermittent PTH during recovery from hindlimb unloading: impact on mechanical properties and bone formation rate in cortical bone." N/A Oct-2004